

Notes on grifting by optimal transport

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Consider an affine measurable space $\mathcal{S} \subset \mathbb{R}^d$. Fixing $(\Omega, \Sigma, \mathbb{P})$, $\Omega = \mathcal{S}$, $\Sigma = \mathcal{B}(\mathcal{S})$ the Borell σ -algebra of $\mathcal{S}$, and $\mathbb{P}$ a probability measure on (Ω, Σ) . Let $v := (v_t)_{t \in \mathbb{N}}$ be a collection of Σ -measurable automorphisms on $\mathcal{S}$. For any random variable $X : \Omega \rightarrow \mathcal{S}$, let $\mathcal{L}(X)$ denote the law of X under $\mathbb{P}$.

The dynamical system of a drifting model (Deng et al., 2026) evolves the prediction X during training according to the following Markov chain:

$$X_{t+1}^v = X_t^v + v_t(X_t) \in \mathcal{S} \quad (1)$$

where $v : (t, x) \in \mathbb{N} \times \mathcal{S} \rightarrow \mathcal{V} \subset \mathcal{S}$. Equivalently, letting μ_0 be a probability measure on (Ω, Σ) , this can be viewed in terms of the evolution of the law of X_t from its initial distribution μ_0 as

$$\begin{cases} \mathcal{L}(X_{t+1}^v) = (\text{Id} + v_t)_\# \mathcal{L}(X_t^v) \\ \mathcal{L}(X_0^v) = \mu_0 \end{cases} \quad (2)$$

The development of drifting models Deng et al. (2026) rests on the idea of crafting (by hand) a vector-field-valued map v which is asymptotically stable and whose attracting equilibrium induces $X_\infty \sim \nu$ for a given target distribution $\nu \in \mathcal{P}(\mathcal{S})$. By using the methodology of least-squares fitting outlined in Deng et al. (2026), the specification of v can be leveraged into the *training* of a neural network f which yields an approximate pushforward map transforming a given noise distribution μ_0 into the target distribution ν , i.e. $\nu \approx f_\# \mu_0$.

From a technical standpoint, the authors of Deng et al. (2026) construct a family of vector fields v depending on ν and $\mathcal{L}(X_t^v)$ which admits as an equilibrium the target distribution ν . However, under general conditions it remains unclear whether the resulting mean-field dynamics of X_t will meaningfully converge to ν as $t \rightarrow \infty$, despite them giving some sufficient conditions.

A contrario, it is tempting to reinterpret (1) as a control problem, and try and determine v through the specification of an objective function to be optimised. At an intuitive level, we can then view the efforts of Deng et al. (2026) as an ad-hoc attempt to guess a control v which will yield the desired target distribution ν .

Inspired by the recent work of Müller et al. (2026), we propose to study this as a *measure constrained* optimal control problem.

1. Discounted setting

Let $\gamma \in [0, 1)$, $c : \mathcal{S}^2 \rightarrow \mathbb{R}$ be a measurable cost function, and let $\mu_0 \in \mathcal{P}(\mathcal{S})$ be a given initial distribution, and consider the optimal control problem,

$$\begin{aligned} J_p : \mu_0 \in \mathcal{P}(\mathcal{S}) \mapsto \min_{v \in \mathcal{V}} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t c(X_t^v, X_{t+1}^v) \right] \\ \text{subject to } X_{t+1}^v = X_t^v + v_t(X_t^v) \text{ for all } t \geq 0, \end{aligned} \quad (3)$$

$$X_0^v \sim \mu_0,$$

where $\mathcal{V}$ is a set of admissible controls. By abuse of notation, we will write $J_p(x)$ for the value of the optimal control problem when $\mu_0 = \delta_x$ is a Dirac measure at $x \in \mathcal{S}$. It is easy to see that a dynamic programming principle holds for J_p , namely

$$J_p(x) = \min_{\bar{v} \in \mathcal{S}} \{c(x, x + \bar{v}) + \gamma J_p(x + \bar{v})\}. \quad (4)$$

and that thus

$$J_p(\mu_0) = \int J_p(x) \mu_0(dx). \quad (5)$$

The optimal policy $v^* \equiv v^*$ is then readily extracted from the minimiser of the right-hand side of the dynamic programming principle.

Consequently, for any x_0 , optimal trajectories of the system (1) follow the constant speed curves

$$t \mapsto \sum_{s=0}^{t-1} v^*(x_s) = (\text{Id} + v^*)^t x_0, \quad (6)$$

where $x_{s+1} = x_s + v^*(x_s)$. This suggests an alternative fitting approach than the one proposed in [Deng et al. \(2026\)](#): find an approximate fixed point of J_p using least-squares and a Neural Network, somehow extract v^* (for example with c being a squared norm to obtain the network gradient), and then predictions can be computed by feeding forward through the network a few times. However the issue is we don't have a good way to ensure $X_t \rightarrow \nu$ from this. But it could give us ideas for after we include the constraint $\mathcal{L}(X_t^v) \rightarrow \nu$ in the optimal control problem.

To reduce the simulation cost, we can also try to find an efficient representation of $(\text{Id} + v^*)^t$ for t either fixed or as an input parameter of a neural network approximating it (this gets us closer in spirit to the simulation-free approach of drifting models). If v^* is a gradient field, then we can use the fact that $(\text{Id} + v^*)^t$ is the gradient of a convex potential field perhaps, by writing this as a gradient flow (this would mirror the approach of [Müller et al. \(2026\)](#)). If v^* is itself a linear operator, then we can directly operate on the spectrum of v^* to compute the exponential for any t very easily. How to justify that v^* is a linear operator is unclear, maybe we can use a latent space / NTK representation or restrict the control class to linear operators. An approximate alternative is presented by linearising v^* around x_0 and then computing the exponential of the linearised operator. This corresponds to doing a single shot straight path simulation. I suppose we could also try to be clever by combining different points like in a Runge-Kutta scheme to get very good approximations on the cheap.

1.1. Formulations of the simulation constraint

The dynamical system (1), in the equilibrium condition specified by [Deng et al. \(2026\)](#), would ideally satisfy

$$\lim_{t \rightarrow \infty} \mathcal{L}(X_t^v) = \nu \text{ i.e. } \mathcal{L}(X_{t+1}^v) \approx \mathcal{L}(X_t^v) \approx \nu, \quad (7)$$

(understood as in distribution) for t sufficiently large, where $\nu \in \mathcal{P}(\mathcal{S})$ is a given target distribution, and equality is in distribution. If we restrict ourselves to distributions with finite variance (sledgehammer argument: images are bounded), by the metrisation of weak convergence by the Wasserstein-2 distance, this is equivalent to

$$\mathcal{W}_2(\mathcal{L}(X_t^v), \nu) \approx \mathcal{W}_2(\mathcal{L}(X_{t+1}^v), \mathcal{L}(X_t^v)) \approx 0. \quad (8)$$

One path: fix as a limit constraint, combine with the other constraint and relax? The stationarity condition can be equivalently rewritten in terms of the velocity field v_t as

$$\sup_{\varphi \in C_b(\mathcal{S})} \int \varphi(x + v_t(x)) - \varphi(x) \mathcal{L}(X_t^v)(dx) = 0, \quad (9)$$

which can be computed using a GAN but this sup is very bad. Anticipating a bit, if $\mu_t^v = \mathcal{L}(X_t^v, X_{t+1}^v)$ and denoting by $\oplus$ the direct sum, this is a convex constraint on μ_t^v :

$$\mu_t^v \in \mathcal{P}_\gamma(\mathcal{S}^2) : \sup_{\varphi \in C_b(\mathcal{S})} \int \varphi \oplus \varphi(x, y) \mu_t^v(dx, dy) = 0. \quad (10)$$

Equivalently the distributional constraint can be rewritten as

$$\sup_{\varphi \in C_b(\mathcal{S})} \int \varphi(x) \nu(dx) - \varphi(x) \mathcal{L}(X_t^v)(dx) = 0. \quad (11)$$

One remark about imposing stationarity, if v is constant then $\mu := \mathcal{L}(X_t^v) = \mathcal{L}(X_{t+1}^v)$ implies that $\mu = (\text{Id} + v)|_\# \mu$, and thus $\mu \in \ker(\text{Id} + (\text{Id} + v)|_\#)$, understood as a linear operator on signed measures. Note that this is a linear constraint on μ but it is coupled to the primal adjoint variable v .

Another way to impose stationarity is to require that $T := \text{Id} + v$ is admissible as a Monge map for the transport problem between μ and itself, i.e. $T|_\# \mu = \mu$. This can perhaps be relaxed to a Kolmogorov coupling constraint on the joint. Come to think of it, this is exactly a rewriting of the constraints that the joint be the joint of the stationary Markov chain with transition T .

1.2. Dual control

While the primal theory tackles (3) head-on, we will instead focus on the (control-theoretic) dual formulation of the problem, which is linear in the space of measures.

To do so, let $\mathcal{M}_+(\mathcal{S}^2)$ be the space of σ -finite positive measures on $\mathcal{S}^2$, and let $\mathcal{P}(\mathcal{S}^2) \subset \mathcal{M}(\mathcal{S}^2)$ be the space of probability measures on $\mathcal{S}^2$. We define the “forward” and “backward” marginalisation operators $\mathcal{F}$ and $\mathcal{B}$ on $\mathcal{M}_+(\mathcal{S}^2)$ as follows:

$$\mathcal{F} : \mu \in \mathcal{M}_+(\mathcal{S}^2) \rightarrow \int_{\mathcal{S}} \mu(dx, \cdot) \in \mathcal{M}_+(\mathcal{S}) \quad (12)$$

$$\mathcal{B} : \mu \in \mathcal{M}_+(\mathcal{S}^2) \rightarrow \int_{\mathcal{S}} \mu(\cdot, dy) \in \mathcal{M}_+(\mathcal{S}) \quad (13)$$

For any $\gamma \in [0, 1)$, let $\mathcal{P}_\gamma(\mathcal{S}^2) := (1 - \gamma)^{-1} \mathcal{P}(\mathcal{S}^2)$ be the subspace of $\mathcal{M}_+(\mathcal{S}^2)$ of measures with total mass $(1 - \gamma)^{-1}$.

For any $v \in \mathcal{V}$, with μ_0 implicitly fixed, let $\mu^v \in \mathcal{P}_\gamma(\mathcal{S}^2)$ be the measure defined by

$$\mu^v(dx, dy) := \sum_{t=0}^{\infty} \gamma^t \mathcal{L}(X_t^v, X_{t+1}^v)(dx, dy). \quad (14)$$

This is equivalently written as

$$\mu^v : (A, B) \in \Sigma^2 \rightarrow \sum_{t=0}^{\infty} \gamma^t \mathbb{P}(X_t^v \in A, X_{t+1}^v \in B) = \sum_{t=0}^{\infty} \gamma^t \mathbb{P}(X_t^v \in A, (\text{Id} + v_t)(X_t^v) \in B). \quad (15)$$

The *control-theoretic dual* is the linear problem

$$J_d := \min_{v \in \mathcal{V}} \int c(x, y) \mu^v(dx, dy) \quad (16)$$

$$\text{subject to } \mu^v \in \mathcal{P}_\gamma(\mathcal{S}^2), \quad \mathcal{B}\mu^v = \mu_0 + \gamma \mathcal{F}\mu^v \quad (17)$$

$$\mathcal{F}\mu^v = \sum_{t=0}^{\infty} \gamma^t \bigcirc_{s=1}^t (\text{Id} + v_s)|_\# \mu_0$$

Lemma 1 *The control-theoretic duality is strong, i.e., $J_p = J_d$. (We will treat minimiser equivalence later.)*

Proof That $\int c d\mu^v = \mathbb{E}[\sum_{t=0}^{\infty} \gamma^t c(X_t^v, X_{t+1}^v)]$ is immediate from the definition of μ^v .

To establish the constraints of the dual, note that, by definition $\mathcal{B}\mu^v = \sum_{t=0}^{\infty} \gamma^t \mathcal{L}(X_t^v)$, and thus

$$\mathcal{F}\mu^v = \sum_{t=0}^{\infty} \gamma^t \mathcal{L}(X_{t+1}^v) = \frac{1}{\gamma} (\mathcal{B}\mu^v - \mu_0). \quad (18)$$

This establishes the constraint associated to $X_0 \sim \mu_0$. The constraint for the dynamics directly leads to

$$\mathcal{F}\mu^v = \sum_{t=0}^{\infty} \gamma^t \mathcal{L}(X_{t+1}^v) = \sum_{t=0}^{\infty} \gamma^t (\text{Id} + v_t)_\# \mathcal{L}(X_t^v) = \sum_{t=0}^{\infty} \gamma^t \bigcirc_{s=1}^t (\text{Id} + v_s)_\# \mu_0. \quad (19)$$

■

1.3. A Remark on minimal energy configurations

If we set $c(x, y) = \|x - y\|^2$, then the optimal control problem (3) becomes

$$\begin{aligned} \min_{v \in \mathcal{V}} \quad & \sum_{t=0}^{\infty} \gamma^t \|v_t\|_2^2 \\ \text{subject to } & X_{t+1}^v = X_t^v + v_t(X_t) \text{ for all } t \geq 0, \\ & X_0 \sim \mu_0, \end{aligned} \quad (20)$$

The constraints vanish in this case, as the objective is independent of the distribution of X_t^v . The optimal control problem then reduces to a standard optimal control problem, and the optimal control is given by $v_t^* \equiv 0$ for all $t \geq 0$.

For this to be useful for the transport problem, we need to balance with the constraints so that the movement is uniform in time rather than concentrated at infinity.

2. Finite Horizon formulations

If we set the problem as finite horizon, it is much easier to lay out the constraint we care about for generation.

In particular, we can consider the following finite-horizon optimal control problem, in continuous time, with a terminal constraint:

$$\begin{aligned} J_p(0, \mu_0) &:= \min_{v \in \mathcal{V}_0} \int_0^1 \int_0^1 c(X_t^v, v_t(X_t^v)) dt d\mu_0 \\ \text{subject to } & \dot{X}_t^v = v_t(X_t^v) \text{ for all } t \in [0, 1], \\ & X_0^v \sim \mu_0, \\ & \mathcal{L}(X_1^v) = \nu \end{aligned} \quad (21)$$

We can then define the cost-to-go functions as $J_p(t, \mu) := \min_{v \in \mathcal{V}_t} \int_t^1 c(X_s^v, v_s(X_s^v)) ds$ subject to $\dot{X}_s^v = v_s(X_s^v)$ for all $s \in [t, 1]$, $X_t^v \sim \mu$, and $\mathcal{L}(X_1^v) = \nu$. In either case $\mathcal{V}_t$ is the set of admissible controls on $[t, 1]$.

When $c(x, v) = \|v\|_2^2$, this is exactly the Benamou-Brenier formulation of optimal transport

$$\mathcal{W}_2^2(\mu_0, \nu) = \min_{v \in \mathcal{V}_0} \int_0^1 \int \|v_t(x)\|_2^2 d\mu_t(x) dt \quad (22)$$

$$\text{subject to } \frac{d}{dt} \mu_t + \nabla \cdot (v_t \mu_t) = 0 \text{ for all } t \in [0, 1], \quad (23)$$

$$\mu_0 = \mu_0,$$

$$\mu_1 = \nu$$

$$t \mapsto \mu_t \text{ a.c. and } \mathcal{P}_2(\mathcal{S}). \quad (24)$$

Discretising the problem as in [Müller et al. \(2026\)](#), we can solve this via a primal-dual approach, including with only sample access to ν .

This forgets about the stationarity completely, but notice that any convex function of v induces a stationary control and thus immediate stationarity of the velocity.

Something that could be interesting is to include in the cost a term such as $\|x\|_1$ to encourage a turnpike behaviour, pushing into a latent space and back out of it. We can also replace the squared norm with $s \|v\|_2^2$ or with $\|v\|_2$ in order to encourage faster convergence to the target distribution from μ_0 .

3. Alternative mean-field control formulations

Another way to think about the drifting model optimisation is to treat the empirical data as a finite-particle approximation of a mean-field control problem. In particular, we can view (3) as the mean-field control problem

$$J_p(\mu_0) := \min_{v \in \mathcal{V}} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t c(X_t^v, \mathcal{L}(X_t^v), v_t) \right] \quad (25)$$

subject to $X_{t+1}^v = X_t^v + v_t(X_t^v)$ for all $t \geq 0$,
 $X_0^v \sim \mu_0$,

where c is now a function of the law of X_t^v and the velocity field v_t . If we take, for instance, $c(x, \mu, v) = \|v\|_2^2 + \mathcal{W}_2^2(\mu, \nu)$, then the optimal control problem becomes

$$J_p(\mu_0) := \min_{v \in \mathcal{V}} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t \|v_t(X_t^v)\|_2^2 + \mathcal{W}_2^2(\mathcal{L}(X_t^v), \nu) \right] \quad (26)$$

subject to $X_{t+1}^v = X_t^v + v_t(X_t^v, \mathcal{L}(X_t^v))$ for all $t \geq 0$,
 $X_0^v \sim \mu_0$,

which is (we can always change the decay scheme of costs to have finite horizon) a penalised version of the Benamou-Brenier transport problem which encourages convergence to the target distribution ν . Of course, the downside of this is that the velocity field will depend on the attraction of all particles to all samples, just as in the original drifting model.

Below is a small generated overview of what we can hope to do in terms of resolution. Basically, everything will work the same way as with non-measure-valued states, but there will be a nastier measure-valued Bellman equation (28) to solve. However, the resulting optimal control is still a feedback control, and there is a rich literature on the resolution using, say JKO-type schemes. The path to samples it to view our algorithm as directly being the numerical scheme on this problem, including a sample collected to represent the two measures μ and ν via propagation of chaos.

Restricting to Markov feedback controls $v_t(\cdot) = \alpha(\cdot, \mu_t)$ with $\mu_t = \mathcal{L}(X_t)$, the induced one-step map is

$$T_\mu^\alpha(x) := x + \alpha(x, \mu), \quad \mu^+ = (T_\mu^\alpha)_\# \mu.$$

The dynamic programming principle on $\mathcal{P}_2(\mathcal{S})$ then takes the form

$$J_p(\mu) = \inf_{\alpha} \left\{ \int_{\mathcal{S}} c(x, \mu, \alpha(x, \mu)) \mu(dx) + \gamma J_p((T_\mu^\alpha)_\# \mu) \right\}. \quad (27)$$

This is the natural discrete-time Bellman equation on the Wasserstein space of probability measures; see for instance [Pham and Wei \(2017\)](#); [Bensoussan et al. \(2013\)](#); [Carmona and Delarue \(2018a\)](#). In particular, the optimal feedback should be characterised by minimisers of the right-hand side of (27), exactly as in the finite-dimensional case, but now after lifting the state variable from x to the whole law μ .

Assuming that J_p is differentiable in the sense of Lions¹, one may define the decoupling field

$$U(\mu, x) := \frac{\delta J_p}{\delta \mu}(\mu)(x),$$

and then formally differentiate (27) with respect to μ . If $\alpha^*(\cdot, \mu)$ is an optimal feedback and $T_\mu^*(x) := x + \alpha^*(x, \mu)$ with $\mu^+ = (T_\mu^*)_\# \mu$, this leads to the discrete-time master equation

$$\begin{aligned} U(\mu, x) = & c(x, \mu, \alpha^*(x, \mu)) + \gamma U(\mu^+, T_\mu^*(x)) + \int_S \frac{\delta c}{\delta \mu}(y, \mu, \alpha^*(y, \mu))(x) \mu(dy) \\ & + \gamma \int_S D_x U(\mu^+, T_\mu^*(y)) \cdot \frac{\delta T_\mu^*}{\delta \mu}(y)(x) \mu(dy), \end{aligned} \quad (28)$$

again at a purely formal level. Equation (28) is the discrete analogue of the continuous-time master equation from mean-field game and mean-field type control theory, and should be understood as a fixed-point equation for the Lions derivative of the value functional; compare with [Cardaliaguet et al. \(2019\)](#); [Carmona and Delarue \(2018b\)](#). It may be possible to formulate this more cleanly by first passing to a finite-horizon problem and then sending the horizon to infinity.

As for solving the problem, the quadratic term $\|v\|_2^2$ by itself does not make the optimisation a simple least-squares problem: the nonlinearity is hidden in the pushforward constraint $\mu^+ = (\text{id} + \alpha(\cdot, \mu))_\# \mu$ and in the dependence of the running cost on the evolving law. Thus a direct calculus-of-variations approach is plausible only in special regimes where one can relax the control problem to a convex optimisation over curves of measures or transport plans, as in Benamou–Brenier or JKO-type formulations; see [Santambrogio \(2015\)](#); [Jordan et al. \(1998\)](#); [Ambrosio et al. \(2005\)](#). In the general mean-field control setting one typically needs either the dynamic programming/master equation route or the stochastic/variational maximum principle developed in [Bensoussan et al. \(2013\)](#); [Carmona and Delarue \(2018a,b\)](#).

4. Transient distribution model

The original drifting model paper [Deng et al. \(2026\)](#) considers a Markov chain over the training time with the goal of following its transition kernel to eventually reach a stationary distribution which is close to or equal to the target distribution ν . In the paper, the transition kernel is constructed by hand, with some conditions which should ensure that the stationary distribution is indeed ν .

In contrast, it is tempting to view the problem as a control problem, and to explicitly optimise the transition kernel to reach the target distribution ν in as few learning steps as possible. This imposes a constraint on the ergodic control problem induced by the controlled Markov chain induced by all possible vector fields v during the learning dynamics. To handle this difficulty, we propose to analyse the resulting problem directly through a Lagrangian relaxation with the goal of performing a primal-dual optimisation over the vector field v in which we will use samples for access to ν .

In practice, much like in the original drifting model paper, we will subsequently use optimise over a parametric family of vector fields as we will parametrise the generative model by a neural network. Because of the chain rule, we will need to compute the gradients of the full Lagrangian with respect to v , which is the paradigm we adopt here to begin with. Note that, using the trick of [Deng et al. \(2026\)](#), we don't need to explicitly differentiate through the network, it is sufficient to find a good v and then minimise the least-squares loss to stationarity, or use the formulation of [Müller et al. \(2026\)](#) and generate directly from samples of μ_0 via the transport v (setting the initial network to the identity).

1. See e.g. [Bao et al. \(2021\)](#), basically it's Fréchet derivative but for measures. We can easily assume densities and work with them instead.

4.1. Lagrangian of the problem

If we let v be a constant velocity field, assuming that v induces a stationary distribution q_v for the Markov chain (1), i.e. $q_v = (\text{Id} + v)|_{\#} q_v$, then we can define the transient distribution of the Markov chain as

$$\phi : A \in \mathcal{B}(\mathcal{S}) \rightarrow \sum_{t=0}^{\infty} \mathbb{P}(X_t^v \in A) - q_v(A) \in [0, \infty]. \quad (29)$$

In this case, we can aim to optimise over v to minimise the energy of the transient distribution, according to some cost criterion. For instance, we can consider the following optimisation problem:

$$\begin{aligned} \min_{v \in \mathcal{S}} \int_{\mathcal{S}} c(x) \phi(dx) \\ \text{subject to } q_v = (\text{Id} + v)|_{\#} q_v. \end{aligned} \quad (30)$$

Remark 2 *It's important to note how different this formulation is from the Value Driven Transport paper of Müller et al. (2026). There is no transport cost from x to $x + v(x)$, here (at least not explicitly). What is penalised is a weighting of the system's convergence rate to the stationary distribution based on different costs in space. If we proceed we should be able to show this effect by making $c = 1 + \mathbb{1}_S$ for a set S of states, such as a halfspace. We should in general take $c \equiv 1$ if we have no prior knowledge of the target distribution ν .*

A simple calculation shows that

$$\phi_v = \mu_0 - q_v + (\text{Id} + v)|_{\#} \phi_v, \quad (32)$$

if and only if ϕ_v is defined as above. We can add this linear constraint to the optimisation problem, and relax the objective to separate ϕ_v and v . This gives

$$\min_{v \in \mathcal{S}, \phi \in \mathcal{M}} \langle c | \phi \rangle \quad (33)$$

$$\text{subject to } q_v = (\text{Id} + v)|_{\#} q_v, \quad (34)$$

$$\phi_v = \mu_0 - q_v + (\text{Id} + v)|_{\#} \phi_v \quad (35)$$

$$\phi \geq \phi_v. \quad (36)$$

The transient distribution ϕ_v also has two properties which we can switch to constraints on ϕ to make optimising over ϕ easier. First, $\phi(\Omega) = 0$, as both $\mathbb{P}(X_t^v \in \cdot)$ and q_v are probability measures. Notice that ϕ is unbounded, as is easily shown by taking the two-state system with transition probability matrix

$$P_v = \begin{pmatrix} 1 - \varepsilon & \varepsilon \\ 0 & 1 \end{pmatrix} \quad (37)$$

wherein the transient part has mass $(1 - \varepsilon)^{-1}$, on s_1 and $1 - (1 - \varepsilon)^{-1}$ on s_2 , while $q(s_2) = 1$ for any $\varepsilon > 0$. Sending $\varepsilon \rightarrow 1$ sends $\phi(s_1) \rightarrow \infty$ and $\phi(s_2) \rightarrow -\infty$.

Adding this, plus the constraint that q_v be the target distribution ν , we can write the optimisation problem as

$$\min_{v \in \mathcal{S}, \phi \in \mathcal{M}} \langle c | \phi \rangle \quad (38)$$

$$\text{subject to } q_v = (\text{Id} + v)|_{\#} q_v, \quad (39)$$

$$\phi_v = \mu_0 - q_v + (\text{Id} + v)|_{\#} \phi_v \quad (40)$$

$$q_v = \nu \quad (41)$$

$$\phi \geq \phi_v, \quad (42)$$

$$\phi(\Omega) = 0. \quad (43)$$

While this system has non-linearities (in v) in the constraints, it doesn't prevent us from writing the primal-dual system and trying to solve it. The main issue for solving is that we need to be able to 1) compute the stationary distribution q_v for any v , 2) compute the transient distribution ϕ_v for any v , and 3) compute the gradient of the transient distribution with respect to v (maybe doable via PG theorem?).

The Lagrangian for the system is

$$\begin{aligned} \mathcal{L}(v, \phi, \lambda_1, \lambda_2, \lambda_3, \lambda_4) = & \langle c | \phi \rangle + \langle \lambda_1 | q_v - (\text{Id} + v) |_{\#} q_v \rangle + \langle \lambda_2 | \phi_v - \mu_0 + q_v - (\text{Id} + v) |_{\#} \phi_v \rangle \\ & + \langle \lambda_3 | q_v - \nu \rangle + \langle \lambda_4 | \phi_v - \phi \rangle + \lambda_5 \phi(\Omega), \end{aligned} \quad (44)$$

or, reordered,

$$\begin{aligned} \mathcal{L}(v, \phi, \lambda_1, \lambda_2, \lambda_3, \lambda_4) = & \langle c - \lambda_4 | \phi \rangle + \lambda_5 \phi(\Omega) + \langle \lambda_3 | \nu \rangle + \langle \lambda_1 + \lambda_2 + \lambda_3 | q_v \rangle + \langle \lambda_2 + \lambda_4 | \phi_v \rangle \\ & - \langle \lambda_1 | P_v q_v \rangle - \langle \lambda_2 | P_v \phi_v \rangle, \end{aligned} \quad (45)$$

which is linear in ϕ and q_v , but non-linear in v .

4.2. Gradients of the Lagrangian

The gradients with respect to the multipliers and ϕ are easy provided knowledge of ϕ_v and q_v (except for λ_3 which requires the target ν but it can take SGD). On the other hand, taking the gradient in the control space yields²

$$\begin{aligned} \nabla_v \mathcal{L}(v, \phi, \lambda_1, \lambda_2, \lambda_3, \lambda_4) = & \nabla_v (\langle \lambda_1 + \lambda_2 + \lambda_3 | q_v \rangle + \langle \lambda_2 + \lambda_4 | \phi_v \rangle) \\ & - \nabla_v (\langle \lambda_1 | P_v q_v \rangle + \langle \lambda_2 | P_v \phi_v \rangle) \end{aligned} \quad (46)$$

where $P_v := (\text{Id} + v) |_{\#}$ is the transition operator of the Markov chain, viewed as a linear operator on signed measures.

We need to be a bit careful about topology here in order to get these derivatives into the inner products. So to be general, let's define a function class $\mathcal{F}$ of functions, which is included in $\mathcal{C}_b^0$ (the pre-dual of $\mathcal{M}$ the space of finite signed measures) and the corresponding $\mathcal{F}$ -weak topology on the space of signed measures which is the smallest topology such that $\mu \mapsto \langle f | \mu \rangle$ is continuous for all $f \in \mathcal{F}$. (In general, we will want to take a dense subset of $\mathcal{C}_b^0$ with nice test-function properties, such as $\mathcal{C}_c^\infty$ or $\mathcal{C}_0^\infty$.)

Definition 3 (Gateaux derivative in the $\mathcal{F}$ -weak topology) *Let $\mathcal{F} \subset \mathcal{C}_b^0(\mathcal{S})$ be a function class, and let $v \mapsto m_v$ be a map from $\mathcal{S}$ to $\mathcal{M}$. We say that m_v is $\mathcal{F}$ -weakly Gateaux differentiable at $v \in \mathcal{S}$ if there exists $D_h m_v : \mathcal{S} \rightarrow \mathcal{M} \in \mathcal{M}$ such that,*

$$\lim_{\varepsilon \rightarrow 0} \sup_{f \in \mathcal{F}} \left| \frac{\langle f | m_{v+\varepsilon h} - m_v \rangle}{\varepsilon} - \langle f | D_h m_v \rangle \right| = 0. \quad (47)$$

We call $D_h m_v$ the $\mathcal{F}$ -weak Gateaux derivative of m_v at v in direction $h \in \mathcal{S}$, and, if $h \mapsto D_h m_v$ is well-defined for all v we call $v \mapsto D_h m_v$ the Gateaux derivative of m_v in direction h .

Similarly, we can define the Gateaux derivative of an operator $v \mapsto O_v$ with values in linear operators on signed measures, by calling $\bar{D}_h O_v$ the Gateaux derivative of O_v at v in direction h if, for all $\mu \in \mathcal{M}$,

$$\lim_{\varepsilon \rightarrow 0} \sup_{f \in \mathcal{F}} \left| \frac{\langle f | O_{v+\varepsilon h} \mu - O_v \mu \rangle}{\varepsilon} - \langle f | \bar{D}_h O_v \mu \rangle \right| = 0. \quad (48)$$

2. For simplicity, we develop below for c independent of v , but the same reasoning applies if c depends on v , up to adding $\langle \nabla_v c | \phi \rangle$.

(We use the notation $\bar{D}$ to distinguish the derivative of an operator applied to a measure-valued map from the derivative of the composed map $v \mapsto O_v m_v$ itself.) Assuming both $v \mapsto m_v$ and $v \mapsto O_v$ are Gateaux differentiable, we can then write the product rule

$$D_h(O_v m_v) = \bar{D}_h O_v m_v + O_v D_h m_v. \quad (49)$$

Thus, we can rewrite (45) as

$$\begin{aligned} \nabla_v \mathcal{L}(v, \phi, \lambda_1, \lambda_2, \lambda_3, \lambda_4) = & \langle \lambda_1 + \lambda_2 + \lambda_3 | D_h q_v \rangle + \langle \lambda_2 + \lambda_4 | D_h \phi_v \rangle \\ & - \langle \lambda_1 | \bar{D}_h P_v q_v + P_v D_h q_v \rangle - \langle \lambda_2 | \bar{D}_h P_v \phi_v + P_v D_h \phi_v \rangle. \end{aligned} \quad (50)$$

Lemma 4 *Let $\mu \in \mathcal{M}$ and $f \in \mathcal{F}$. Assume $\mathcal{F} \subset \mathcal{C}^1(\mathcal{S})$ and then, for any $h \in \mathcal{S}$ and $v \in \mathcal{S}$, the duality pairing of f and the Gateaux derivative of the pushforward operator P_v is given by*

$$\langle f | \bar{D}_h P_v \mu \rangle = \int \nabla f(x + v(x)) \cdot h(x) \mu(dx). \quad (51)$$

Which, conceptually can be understood as operator on signed measures whose adjoint is $(\bar{D}_h P_v)^* : f \mapsto h \nabla f(\cdot + v(\cdot))$, where the gradient is of course only on the argument of f (is this what kids these days call a stopgradient?).

Consequently, $v \mapsto \langle f | P_v \mu \rangle$ is Fréchet differentiable in the $\mathcal{F}$ -weak topology, with Fréchet derivative

$$h \mapsto \int h(x) \nabla f(x + v(x)) \mu(dx). \quad (52)$$

Proof This follows by a Taylor expansion of f and the definition of the pushforward operator:

$$\langle f | P_{v+\varepsilon h} \mu \rangle = \int f(x + v(x) + \varepsilon h(x)) \mu(dx) \quad (53)$$

$$= \int f(x + v(x)) \mu(dx) + \varepsilon \int \nabla f(x + v(x)) \cdot h(x) \mu(dx) + o(\varepsilon), \quad (54)$$

and thus

$$\langle f | \bar{D}_h P_v \mu \rangle = \lim_{\varepsilon \rightarrow 0} \frac{\langle f | P_{v+\varepsilon h} \mu - P_v \mu \rangle}{\varepsilon} = \int \nabla f(x + v(x)) \cdot h(x) \mu(dx). \quad (55)$$

■

While for the $P_v \mu$ terms in the lagrangian ($\mu \in \{q_v, \phi_v\}$), it was convenient to differentiate the inner product with a test function $f \in \mathcal{F}$ and directly express the differential of the pushforward operator in terms of the adjoint action on f , when differentiating the ergodic terms (q_v and ϕ_v), we will need to face the measures themselves, and thus we need to compute the Gateaux derivatives $D_h q_v$ and $D_h \phi_v$, which we do in Lemma 7 and Lemma 10 below.

Before we move onto these rigorous constructions, let us recall that we care about differentiative $v \mapsto \langle f | \mu \rangle$, in other words, we want to prove a Policy Gradient Theorem of the following form.

Theorem 5 *Let q_v be the stationary distribution associated with the operator P_v . Let $f \in \mathcal{F}$ be a test function, then $v \mapsto \langle f | q_v \rangle$ is $\mathcal{F}$ -weakly Gateaux differentiable at v , and its Gateaux derivative in direction $h \in \mathcal{S}$ is given by*

$$D_h \langle f | q_v \rangle = \int \nabla u_f(x + v(x)) \cdot h(x) q_v(dx), \quad (56)$$

where u_f is a solution to the Poisson equation

$$u_f - P_v^* u_f = f - \langle f | q_v \rangle. \quad (57)$$

Proof Under their technical assumptions, the result follows directly from Lemma 7 and Lemma 8. $\blacksquare$

Remark 6 Most often, policy gradient theorems are expressed in terms of the derivative of the score (log-transform of the transition density) of a softmax policy, this will generally not be possible in our setting, as P_v is a transport operator.

Let us now compute the Gateaux derivative of the stationary distribution q_v with respect to v to prove Theorem 5.

Lemma 7 Let q_v be the stationary distribution associated with the operator P_v . Assume that for all sufficiently small $\varepsilon > 0$, $(v + \varepsilon h \in \mathcal{S})$, the operator $(P_{v+\varepsilon h})$ is a Markov operator on probabilities, and it admits a stationary distribution $(q_{v+\varepsilon h})$.³

Assume that $v \mapsto q_v$ is $\mathcal{F}$ -weakly Gateaux differentiable at v , and that $(\text{Id} - P_v)$ admits an inverse on the subspace of zero-mass signed measures.

Then, for any $h \in \mathcal{S}$, the Gateaux derivative of q_v in direction h is the unique zero-mass signed measure m solving

$$(\text{Id} - P_v)m = \bar{D}_h P_v q_v, \quad (58)$$

(understood in the $\mathcal{F}$ -weak topology, still), that is,

$$D_h q_v = (\text{Id} - P_v)^{-1} \bar{D}_h P_v q_v. \quad (59)$$

If, in addition, the Neumann series converges on zero-mass signed measures, then

$$D_h q_v = \sum_{t=0}^{\infty} P_v^t \bar{D}_h P_v q_v. \quad (60)$$

Proof Since q_v is stationary, it satisfies

$$q_v = P_v q_v. \quad (61)$$

Differentiating this identity in the direction h and applying the product rule established in (49) gives

$$D_h q_v = D_h (P_v q_v) = \bar{D}_h P_v q_v + P_v D_h q_v. \quad (62)$$

Rearranging yields

$$(\text{Id} - P_v) D_h q_v = \bar{D}_h P_v q_v. \quad (63)$$

Moreover, since $q_{v+\varepsilon h}$ is a probability measure for every sufficiently small ε , we also have

$$D_h q_v(\Omega) = \lim_{\varepsilon \rightarrow 0} \frac{q_{v+\varepsilon h}(\Omega) - q_v(\Omega)}{\varepsilon} = 0. \quad (64)$$

Thus $D_h q_v$ belongs to the zero-mass subspace, on which $(\text{Id} - P_v)$ is invertible by assumption, so

$$D_h q_v = (\text{Id} - P_v)^{-1} \bar{D}_h P_v q_v. \quad (65)$$

If the Neumann series for $(\text{Id} - P_v)^{-1}$ converges on that subspace, then

$$(\text{Id} - P_v)^{-1} = \sum_{t=0}^{\infty} P_v^t, \quad (66)$$

3. We need to be a bit careful about verifying these assumptions with stability properties.

and therefore

$$D_h q_v = \sum_{t=0}^{\infty} P_v^t \bar{D}_h P_v q_v. \quad (67)$$

■

Finally, let's rewrite the duality bracket $\langle f | D_h q_v \rangle$ in terms of the solution to the Poisson equation, which will give us the form of the policy gradient theorem.

Lemma 8 *Let $f \in \mathcal{F}$, and assume that the Poisson equation*

$$u_f - P_v^* u_f = f - \langle f | q_v \rangle \quad (68)$$

admits a solution $u_f \in \mathcal{C}^1(\mathcal{S}) \cap \mathcal{F}$. Then, for every direction $h \in \mathcal{S}$,

$$D_h \langle f | q_v \rangle = \langle u_f | \bar{D}_h P_v q_v \rangle. \quad (69)$$

Consequently,

$$D_h \langle f | q_v \rangle = \int \nabla u_f(x + v(x)) \cdot h(x) q_v(dx). \quad (70)$$

Proof Since $D_h q_v$ has zero total mass by Lemma 7, we have

$$D_h \langle f | q_v \rangle = \langle f | D_h q_v \rangle = \langle f - \langle f | q_v \rangle | D_h q_v \rangle. \quad (71)$$

Using the Poisson equation, this becomes

$$D_h \langle f | q_v \rangle = \langle (\text{Id} - P_v^*) u_f | D_h q_v \rangle = \langle u_f | (\text{Id} - P_v) D_h q_v \rangle. \quad (72)$$

Applying Lemma 7 yields

$$D_h \langle f | q_v \rangle = \langle u_f | \bar{D}_h P_v q_v \rangle. \quad (73)$$

The final identity then follows from Lemma 4:

$$\langle u_f | \bar{D}_h P_v q_v \rangle = \int \nabla u_f(x + v(x)) \cdot h(x) q_v(dx). \quad (74)$$

■

Remark 9 *The existence of a $\mathcal{C}^1$ solution to the Poisson equation for any sufficiently regular source term f is a regularity theory result for the associated integral equation. It should be well-studied but will require assumptions.*

We can proceed in much the same way for the transient distribution ϕ_v . However, its implicit dependence on q_v (see (32)) will reintroduce the derivative of q_v to the calculations. This produces a second coupled Poisson equation on the advantage function.

Lemma 10 *Assume that $v \mapsto \phi_v$ is $\mathcal{F}$ -weakly Gateaux differentiable at v , and that $(\text{Id} - P_v)$ admits an inverse on the subspace of zero-mass signed measures. Then, for any $h \in \mathcal{S}$, the Gateaux derivative of ϕ_v in direction h is the unique zero-mass signed measure solving*

$$(\text{Id} - P_v) D_h \phi_v = -D_h q_v + \bar{D}_h P_v \phi_v, \quad (75)$$

that is,

$$D_h \phi_v = (\text{Id} - P_v)^{-1} (-D_h q_v + \bar{D}_h P_v \phi_v). \quad (76)$$

Proof Differentiating the fixed-point identity

$$\phi_v = \mu_0 - q_v + P_v \phi_v \quad (77)$$

in the direction h and using (49) gives

$$D_h \phi_v = -D_h q_v + \bar{D}_h P_v \phi_v + P_v D_h \phi_v. \quad (78)$$

Hence

$$(\text{Id} - P_v) D_h \phi_v = -D_h q_v + \bar{D}_h P_v \phi_v. \quad (79)$$

Since $\phi_v(\Omega) = 0$ for every admissible v , we also have $D_h \phi_v(\Omega) = 0$, so $D_h \phi_v$ belongs to the zero-mass subspace. The claimed representation follows by inverting $(\text{Id} - P_v)$ on that subspace. $\blacksquare$

Theorem 11 (Transient measure gradient theorem) *Let $f \in \mathcal{F}$, and assume that the Poisson equations*

$$u_f - P_v^* u_f = f - \langle f | q_v \rangle, \quad (80)$$

$$w_f - P_v^* w_f = u_f - \langle u_f | q_v \rangle \quad (81)$$

admit solutions $u_f, w_f \in \mathcal{C}^1(\mathcal{S}) \cap \mathcal{F}$. Then the map $v \mapsto \langle f | \phi_v \rangle$ is $\mathcal{F}$ -weakly Gateaux differentiable at v , and for every direction $h \in \mathcal{S}$,

$$D_h \langle f | \phi_v \rangle = \langle u_f | \bar{D}_h P_v \phi_v \rangle - \langle w_f | \bar{D}_h P_v q_v \rangle. \quad (82)$$

Consequently,

$$D_h \langle f | \phi_v \rangle = \int \nabla u_f(x + v(x)) \cdot h(x) \phi_v(dx) - \int \nabla w_f(x + v(x)) \cdot h(x) q_v(dx). \quad (83)$$

Proof Since $\phi_v(\Omega) = 0$ for every admissible v , the derivative also has zero total mass, and hence

$$D_h \langle f | \phi_v \rangle = \langle f | D_h \phi_v \rangle = \langle f - \langle f | q_v \rangle | D_h \phi_v \rangle. \quad (84)$$

Using the Poisson equation for u_f gives

$$D_h \langle f | \phi_v \rangle = \langle (\text{Id} - P_v^*) u_f | D_h \phi_v \rangle = \langle u_f | (\text{Id} - P_v) D_h \phi_v \rangle. \quad (85)$$

Applying Lemma 10 yields

$$D_h \langle f | \phi_v \rangle = -\langle u_f | D_h q_v \rangle + \langle u_f | \bar{D}_h P_v \phi_v \rangle. \quad (86)$$

Since $D_h q_v(\Omega) = 0$, we may subtract the constant $\langle u_f | q_v \rangle$ and use the Poisson equation for w_f to write

$$\langle u_f | D_h q_v \rangle = \langle u_f - \langle u_f | q_v \rangle | D_h q_v \rangle \quad (87)$$

$$= \langle (\text{Id} - P_v^*) w_f | D_h q_v \rangle \quad (88)$$

$$= \langle w_f | (\text{Id} - P_v) D_h q_v \rangle. \quad (89)$$

Using Lemma 7 then gives

$$\langle u_f | D_h q_v \rangle = \langle w_f | \bar{D}_h P_v q_v \rangle. \quad (90)$$

Substituting back into (86), we obtain

$$D_h \langle f | \phi_v \rangle = \langle u_f | \bar{D}_h P_v \phi_v \rangle - \langle w_f | \bar{D}_h P_v q_v \rangle. \quad (91)$$

The integral representation follows from Lemma 4 applied to both terms. $\blacksquare$

We can now substitute the stationary and transient policy gradient formulas into the directional derivative of the Lagrangian.

Lemma 12 *Let*

$$f_q := (\text{Id} - P_v^*)\lambda_1 + \lambda_2 + \lambda_3, \quad (92)$$

$$f_\phi := (\text{Id} - P_v^*)\lambda_2 + \lambda_4. \quad (93)$$

Assume that the Poisson equations

$$U_q - P_v^* U_q = f_q - \langle f_q | q_v \rangle, \quad (94)$$

$$U_\phi - P_v^* U_\phi = f_\phi - \langle f_\phi | q_v \rangle, \quad (95)$$

$$W_\phi - P_v^* W_\phi = U_\phi - \langle U_\phi | q_v \rangle \quad (96)$$

admit solutions in $\mathcal{C}^1(\mathcal{S}) \cap \mathcal{F}$. Then the directional derivative of the Lagrangian in the direction $h \in \mathcal{S}$ is given by

$$\begin{aligned} D_h \mathcal{L}(v, \phi, \lambda_1, \lambda_2, \lambda_3, \lambda_4) &= \langle U_\phi - \lambda_2 | \bar{D}_h P_v \phi_v \rangle \\ &\quad + \langle U_q - W_\phi - \lambda_1 | \bar{D}_h P_v q_v \rangle. \end{aligned} \quad (97)$$

Equivalently,

$$\begin{aligned} D_h \mathcal{L}(v, \phi, \lambda_1, \lambda_2, \lambda_3, \lambda_4) &= \int \nabla(U_\phi - \lambda_2)(x + v(x)) \cdot h(x) \phi_v(dx) \\ &\quad + \int \nabla(U_q - W_\phi - \lambda_1)(x + v(x)) \cdot h(x) q_v(dx). \end{aligned} \quad (98)$$

In other words, when this expression is well-defined, the Fréchet derivative of the Lagrangian in the $\mathcal{F}$ -weak topology is given by

$$\nabla_v \mathcal{L}(v, \phi, \lambda_1, \lambda_2, \lambda_3, \lambda_4) = P_v^* \nabla(U_\phi - \lambda_2) d\phi_v + P_v^* \nabla(U_q - W_\phi - \lambda_1) dq_v. \quad (99)$$

Proof Starting from the directional derivative formula in (50) and moving P_v to the test-function side, we obtain

$$\begin{aligned} D_h \mathcal{L}(v, \phi, \lambda_1, \lambda_2, \lambda_3, \lambda_4) &= \langle (\text{Id} - P_v^*)\lambda_1 + \lambda_2 + \lambda_3 | D_h q_v \rangle \\ &\quad + \langle (\text{Id} - P_v^*)\lambda_2 + \lambda_4 | D_h \phi_v \rangle \\ &\quad - \langle \lambda_1 | \bar{D}_h P_v q_v \rangle - \langle \lambda_2 | \bar{D}_h P_v \phi_v \rangle. \end{aligned} \quad (100)$$

By definition of f_q and f_ϕ , this becomes

$$\begin{aligned} D_h \mathcal{L}(v, \phi, \lambda_1, \lambda_2, \lambda_3, \lambda_4) &= D_h \langle f_q | q_v \rangle + D_h \langle f_\phi | \phi_v \rangle \\ &\quad - \langle \lambda_1 | \bar{D}_h P_v q_v \rangle - \langle \lambda_2 | \bar{D}_h P_v \phi_v \rangle. \end{aligned} \quad (101)$$

Applying Theorems 5 and 11 gives

$$\begin{aligned} D_h \mathcal{L}(v, \phi, \lambda_1, \lambda_2, \lambda_3, \lambda_4) &= \langle U_q | \bar{D}_h P_v q_v \rangle + \langle U_\phi | \bar{D}_h P_v \phi_v \rangle \\ &\quad - \langle W_\phi | \bar{D}_h P_v q_v \rangle - \langle \lambda_1 | \bar{D}_h P_v q_v \rangle - \langle \lambda_2 | \bar{D}_h P_v \phi_v \rangle, \end{aligned} \quad (102)$$

and regrouping the terms yields the first formula. The second formula follows from Lemma 4. ■

4.3. Conclusion

We can attack this problem using a primal-dual method, updates in λ_i , $i \in [5]$ are straightforward, while updating v requires solving three Poisson equations, which we might be able to speed up by trying to learn the ergodic operator $(\text{Id} - P_v^*)^{-1}$ as a function of v (or equivalently the solution to the Poisson equation as a function of (f, v)).

The updates depend on ν only for the λ_3 update, unless we decide to collapse $q_v = \nu$ into other constraints but I don't think that will be a good idea. This means we would take λ_3 to belong to a regular function class whose regularity informs the expected regularity of the target distribution ν , then we take a bunch of SGD steps on λ_3 to learn the target distribution. We take some gradient steps in λ_i , $i \neq 3$, compute the solutions to the Poisson equations, and then take a gradient step (or some stochastic/approximate gradient steps) in v to minimise the Lagrangian.

The problem with this is that each step is probably going to be very expensive due to the integral equation resolution. On the upside, we can use a GPU friendly solver or some clever heuristics based on transport equations like in the previous work. One can hope that the convergence of the primal-dual method is very fast in terms of training, akin to a sort of second-order method, since the Poisson equations should be some kind of gradient equations for the ergodic operator $(\text{Id} - P_v^*)^{-1}$, which is the inverse of the Hessian of the Lagrangian in v . It would also be possible to take SGD steps for optimising v by sending individual sample points of μ_0 into q_v by multi-step transport (or even single step?)

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